

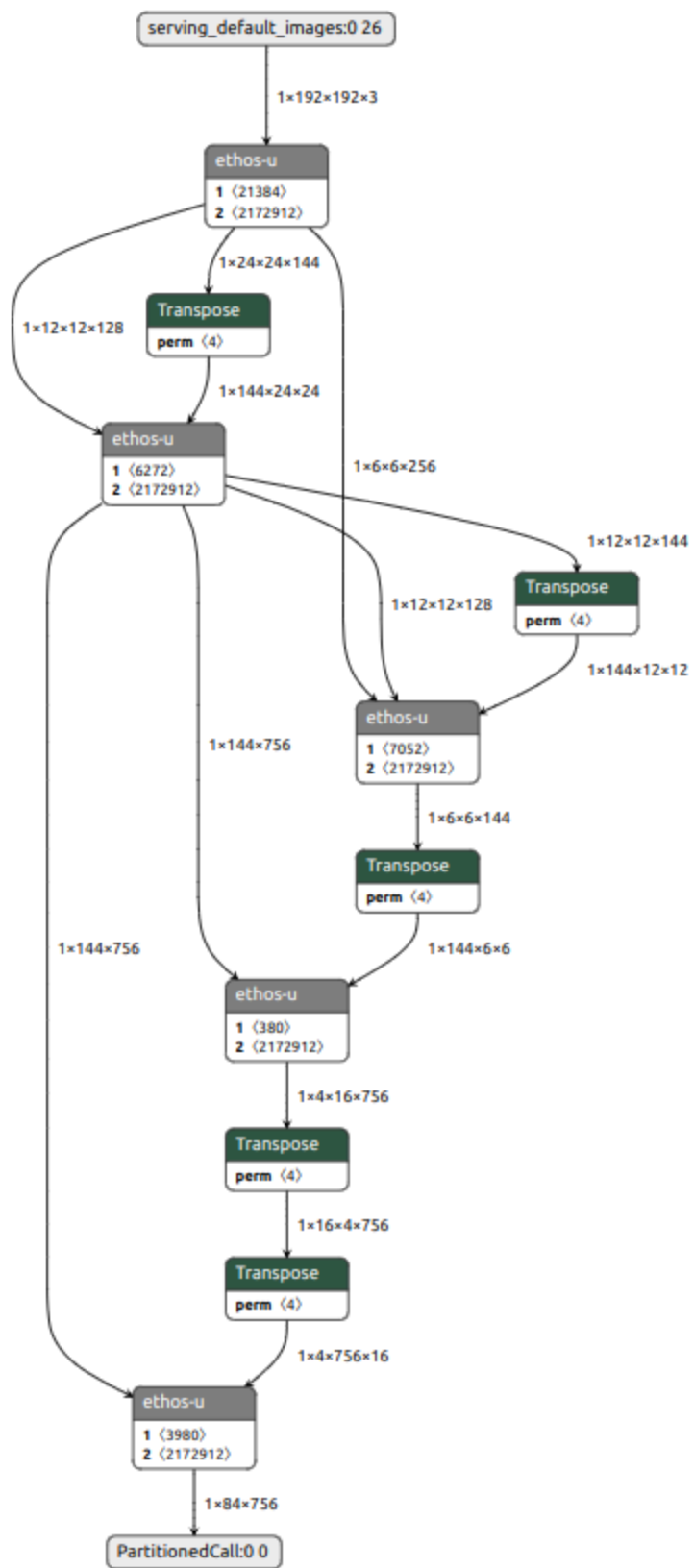
Taken from:

(https://github.com/HimaxWiseEyePlus/YOLOv8_on_WE2?tab=readme-ov-file#export-yolov8n-object-detection-pytorch-model-to-int8-tflite-and-delete-4-transpose-ops)

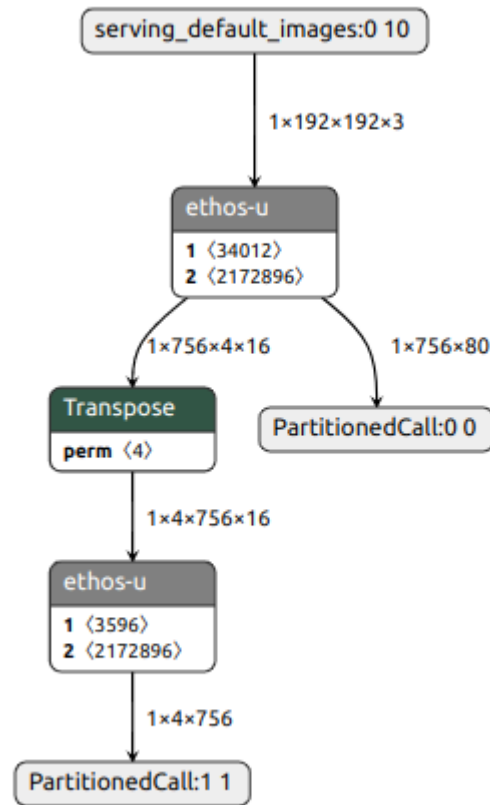
- We give the example here, `replace_192_80cls_transpose_op.json`, but there are something should be changed under following condition.
 1. Be careful, the `op_name` and `param_name` should be changed if needed, depending on the `op_name` and `param_name` of your converted `onnx file`.
 2. If your output class is not 80, you should modify `replace_192_80cls_transpose_op.json` to replace 144 value to `64+{class_num}`. For example, `class_num` is 1, you should modify `replace_192_80cls_transpose_op.json` to replace 144 value to 65, and then it could be correct.

Finally, you can delete 4 `transpose op`, at final int8 tflite.

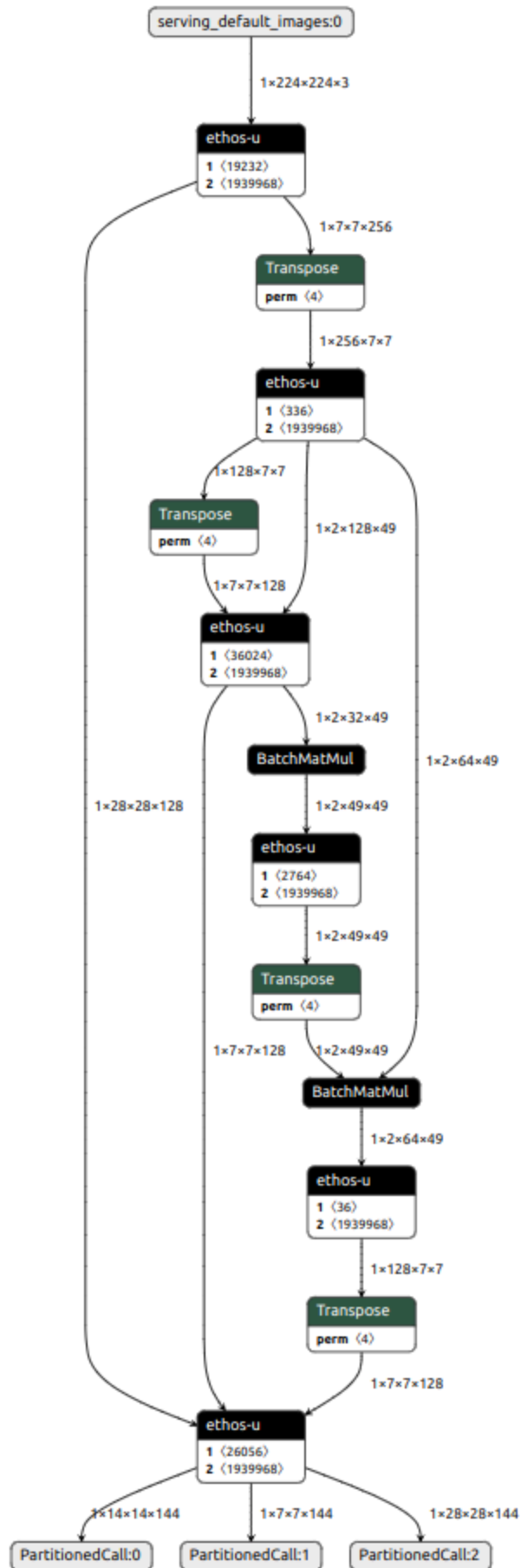
Original tflite which pass vela (YoloV11):



- delete 4 transpose tflite which pass vela (YoloV8)



The output int8 vela tflite model (YoloV11)



The proposed solution provided by Himax, will look to generally follow the same route.

Model Comparison Table

Model	size (pixels)	mAP ^{val} 50-95	Speed CPU ONNX (ms)	Speed T4 TensorRT10 (ms)	params (M)	FLOPs (B)
YOLOv8n	640	37.3	80.4	1.47	3.2	8.7
YOLOv8s	640	44.9	128.4	2.66	11.2	28.6
YOLOv8m	640	50.2	234.7	5.86	25.9	78.9
YOLOv8l	640	52.9	375.2	9.06	43.7	165.2
YOLOv8x	640	53.9	479.1	14.37	68.2	257.8
YOLO11n	640	39.5	56.1	1.5	2.6	6.5
YOLO11s	640	47.0	90.0	2.5	9.4	21.5
YOLO11m	640	51.5	183.2	4.7	20.1	68.0
YOLO11l	640	53.4	238.6	6.2	25.3	86.9
YOLO11x	640	54.7	462.8	11.3	56.9	194.9

- <https://docs.ultralytics.com/compare/yolov8-vs-yolo11/#yolo11-overview>

It's a mature and well-supported model ideal for general-purpose object detection needs. YOLO11, on the other hand, is engineered for speed optimization, making it a superior choice when inference time is critical, especially in CPU-intensive or edge computing scenarios. For applications demanding the fastest possible real-time performance with competitive accuracy, YOLO11 is the preferred option.

Our preference would lie with using YoloV11, as seen from the above and from prior research. But due to time constraints and unknown issues arising with using YoloV11, we are opting to use YoloV8.